Pandas & Scikit-Learn Investment Analysis

GROUP | 14
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SUSTAINABILITY / COMPLEXITY

Milestone One: **Pandas** shows a **stronger overall activity**, but **Scikit-learn's PRs** are more **consistently labeled**, suggesting a **tighter workflow despite its smaller scale.**

Milestone Two: **Pandas merges faster**, but **closes** far more **PRs without merging** raising the question of whether **pandas** has **high standards or is filtering out outside contributions**.

Contributor growth has **slowed** for both projects **since 2020**, making the **"filtering out contributions" theory** worth digging into further.

REASERCH QUESTIONS

1. How long does it take for a pull request to go from submission to merged or closed?
2. How has the number of active contributors changed over time, and what is the ratio of new to returning contributors?
3. Does reviewer concentration (the degree to which reviews are dominated by a small number of contributors) correlate with lower review coverage (PRs merged with reviews), and what does this indicate about the sustainability of each project?

MEDIAN TIME TO MERGE

Pandas
1.0d
Fastest of the two

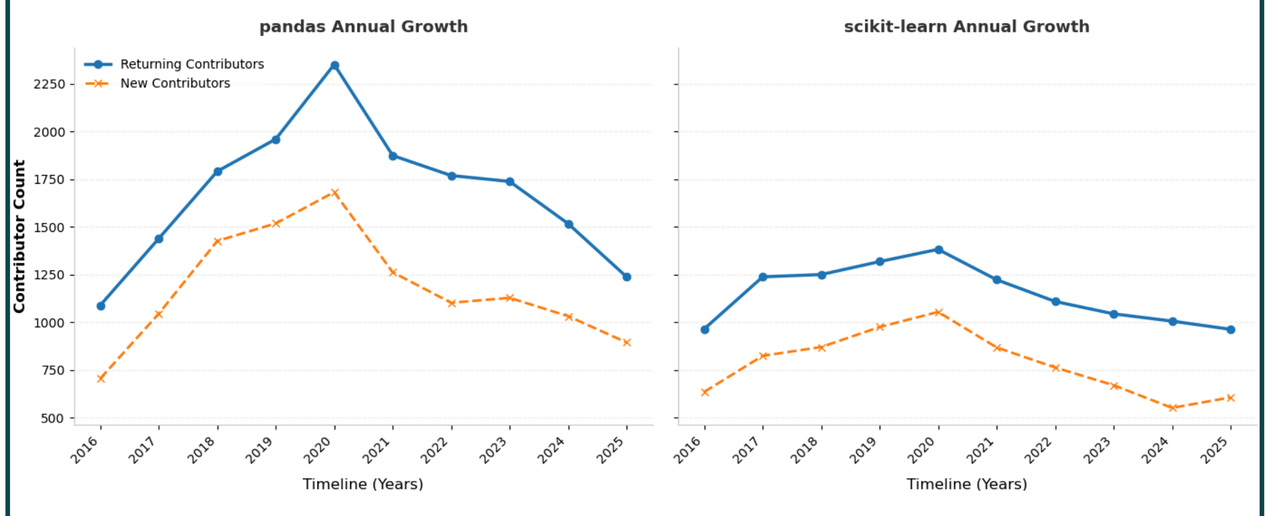
Scikit-learn
1.9d
Slower & lower close-without-merge rate

PRS CLOSED WITHOUT MERGING

Pandas
304x
First sustainability sign

Scikit-learn
38x
Contributors have more meaningful engagement

CONTRIBUTOR HEALTH

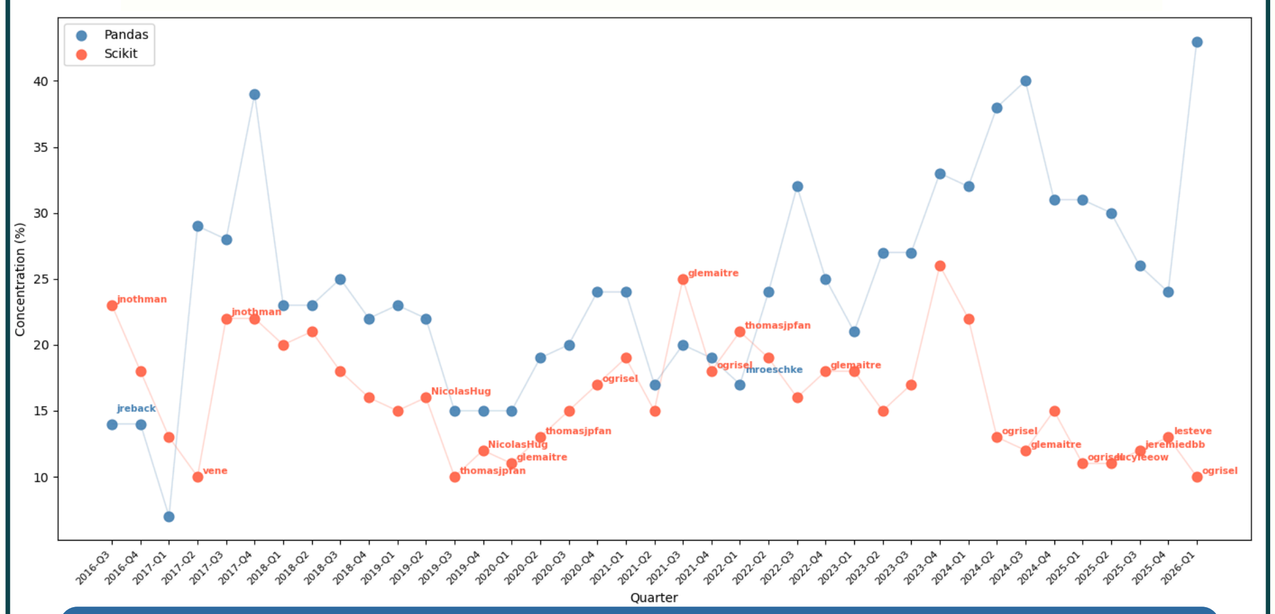


Both curves point downwards, but who depends on fewer to keep functioning

Pandas bus factor
8
Low bus factor exposes key-person dependency.

Scikit-learn bus factor
12+
Bus factor indicates more decentralization.

REVIEW CONCENTRATION

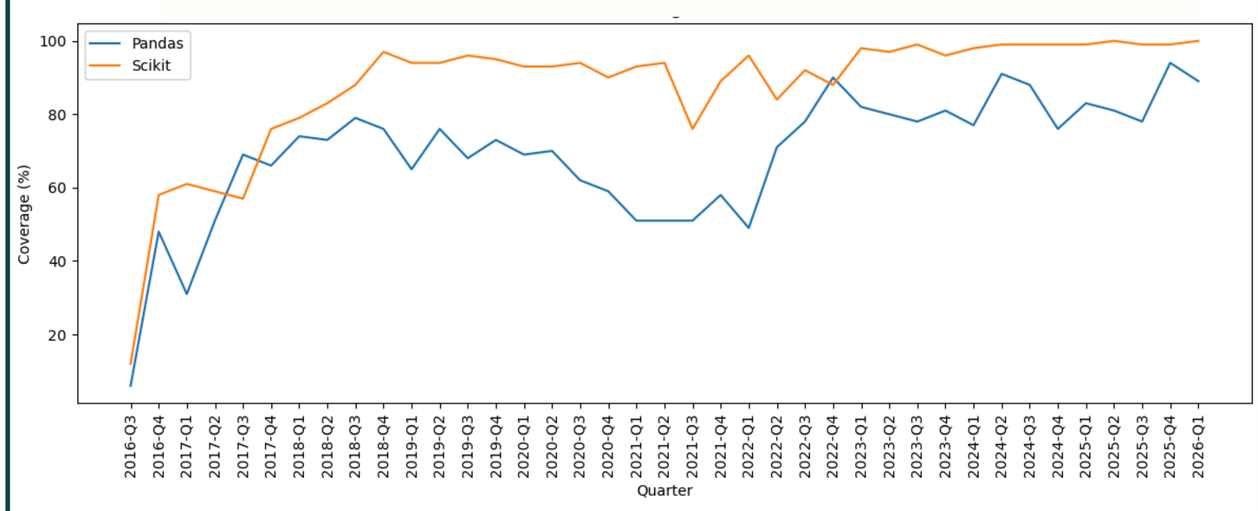


Projects are moving in different directions, but where does responsibility lie

Pandas top-reviewers
30-40%
jreback & mroeschke.

Scikit-learn top-reviewers
10-25%
Rotation between ogrisel, glemaitre & thomasjpfan.

REVIEW COVERAGE



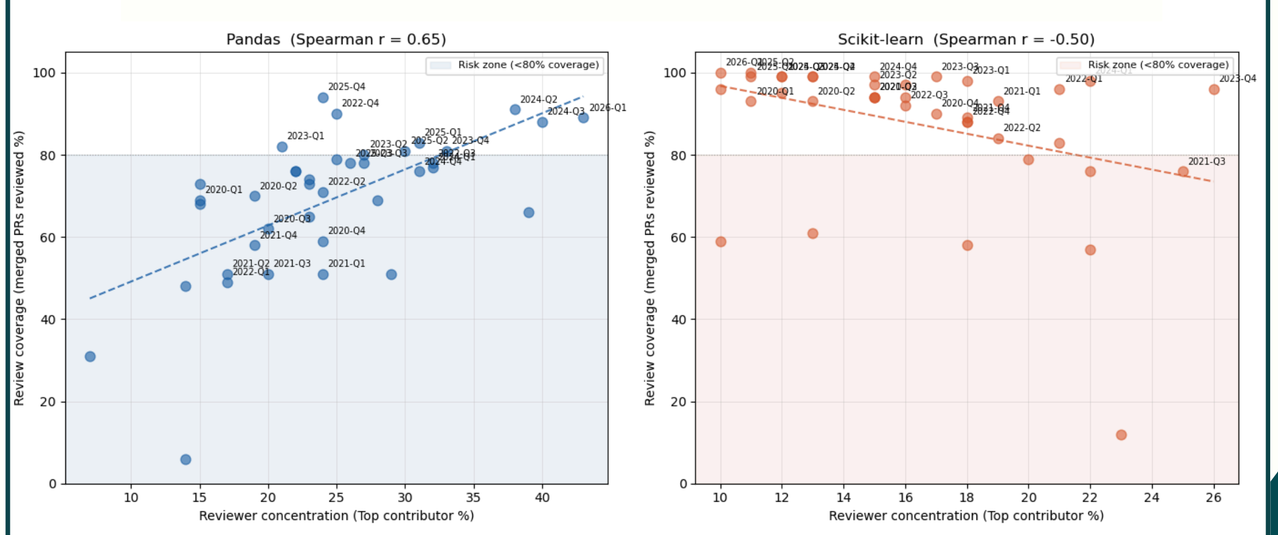
Pandas

Coverage **oscilliates** between **50-80%**, up-tick suggest better quality control in recent years.

Scikit-learn

Coverage is consistently **higher** and far more **stable** since 2018 holding above **80% or above**.

CONCENTRATION VS COVERAGE



Pandas - Spearman R = 0.65

Positive correlation is not good. During earlier years metrics where low, however in recent years **concentration** has reached **risky levels**, while **coverage stagnated**.

Scikit-learn - Spearman r = -0.50

Has **expected relationship**: When **one reviewer dominates coverage suffers**. Even so, its worst recent quarters barely dip bellow 80%

INVESTMENT RECCOMENDATION

Scikit-learn should receive investment since it has;

- better distributed review load
- high coverage
- stable contributor base

which suggests a more sustainable project based on CHAOSS metrics.

STRENGHT AND LIMITATIONS

- Complexity differs, longer PRs or extra reviews may reflect harder problems, not worse process.
- Small/low-risk may need fewer reviews (reveals the real coverage between projects)
+ Centers on contributiors and bus factor rather than surface-level activity counts.
+ Goes past the first impression, analysis differs from where data initially pointed.
+ Concentration and coverage span many variables, capturing a wider slice of project health.

NEXT STEPS

- Cross-reference bug-report rates against review-coverage dips to test whether this is structural risk or proven harm.
- Layer in PR labels to separate complexity-driven duration from process-driven delay.